Safwan Paleri

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Aberdeen, Scotland, United Kingdom

SUMMARY

Professional Game & XR Developer with 3+ years of experience specializing in immersive VR/AR solutions and interactive software. Currently driving technical development at First Competence Ltd, where I architect VR and desktop applications for high-stakes industrial sectors, including Oil & Gas and Energy. Master's graduate with Merit in Computer Games Technology from Abertay University, Scotland, UK, and a Bachelor's degree in Computer Science Engineering (UK 2:1 equivalent) from LPU, India. I have worked on more than 20 projects across 2D, 2.5D, 3D, AR, XR, and VR platforms for Android, iOS, Oculus, standalone, and WebGL, including several developed entirely from scratch, from initial software design to deployment. An avid gamer and passionate game enthusiast, I thrive on creating engaging and immersive interactive experiences.

EXPERIENCE

VR Software Engineer

First Competence ltd (February 2026 - present)

- Architecting and developing high-quality VR simulation products for global leaders in the energy, oil, and gas industry, driving improvements in operational efficiency and safety.
- Transforming immersive game engine technology into specialized tools for complex industrial applications, solving critical training and operational challenges.
- Engineering scalable VR solutions by optimizing rendering pipelines and system architecture to ensure high performance in mission-critical environments.
- Collaborating with cross-functional teams to deliver custom software products that reduce risk and streamline workflow for large-scale industrial projects.

Retail Assistant

Primark (October 2024 – February 2026)

- Operated tills and provided customer service in a high-volume retail environment.
- Handled stock replenishment and floor organization to maintain store presentation.

Unity Game Developer

App Mechanic (August 2022 – December 2024)

- **Gameplay & AI Systems**: Built 2D and 3D gameplay features including combat mechanics, health/reward systems, car customisation, and AI agents (FSM-based NPC enemies, auto-player bots, and LLM-integrated NPC dialogue).
- **AR/VR/XR Development**: Delivered multiple AR/VR projects using Vuforia, Unity AR Foundation, and Niantic Lightship, including GPS-based location puzzles, shared AR object placement across devices, and VR terrain generation from GIS data.
- **Networking & Multiplayer:** Implemented Photon-based multiplayer systems with chat and friend lists, plus fallback AI agents for matchmaking. Developed backend integrations using PlayFab, AWS S3, and Firebase (databases, storage, runtime asset fetching) using .NET and respective APIs.
- **Tools & Systems Development**: Created reusable tools, including a music-driven gameplay system that synchronises player actions with beats, and Unity Addressables pipelines for asset management.

- **Live Ops & Monetization**: Integrated Unity Ads (banner, interstitial, rewarded) and in-app purchase systems across Android and iOS platforms.
- **Generative AI Integration**: Experimented with cutting-edge LLM integrations (ChatGPT/DALL·E for dialogue & image generation, Whisper for speech-to-text, Inworld AI for dynamic NPC dialogue).

Projects Worked Play store links of worked games

Unity Game Developer Intern

Tentaxaur Studios (May 2021 – July 2021)

- Built core gameplay mechanics and interactive features for multiple prototype projects.
- Translated initial design concepts into functional prototypes, enabling the team to visualise and refine gameplay ideas early in development.
- Implemented AI behaviours for NPC enemies and a final boss using finite state machines (FSMs) to handle combat states and conditional transitions.
- Collaborated with designers to test and iterate on mechanics, improving communication between design and programming.

Game Tester Intern

Shahwebsetters (March 2021 - March 2021)

- Conducted gameplay testing focused on mechanics such as scoring systems, timing, and overall balance.
- Evaluated UI elements, including buttons, menus, and settings, to ensure usability and functionality.
- Created and maintained structured bug/issue tracking reports in Google Sheets to support developer iteration

EDUCATION

Master's degree

MSc. Computer Games Technology

Abertay University, Scotland, UK (September 2024 – August 2025)

Bachelor's degree

Bachelor of Technology. Computer Science Engineering (Honours)

Lovely Professional University, Punjab, India (July 2019 – July 2023)

PROJECTS

1. **VR/XR Industrial Safety & Competence Simulations (Unity, C#):** Engineered immersive VR/XR high-risk industrial safety training simulations (Confined Space Entry, Rigging & Lifting, and Working at Height) aligned with OSHA and OPITO standards. Integrated spatial interaction, 3D equipment inspection, step-by-step procedural workflows, and interactive hazard identification in virtual environments. Developed user authentication with local state persistence (7-day checkpoint save/resume systems) and an automated PDF certification pipeline emailing results directly from the standalone VR headset. Implemented core gameplay logic, state machines, and interactive physics mechanics while collaborating closely with UI/UX and 3D art teams.
   VR Show Reel - Rigging and Lifting , VR Show Reel - Confined Space
2. **LLM Game Dialogue & Generative AI Evaluation Tool (Unity, C#):** Architected a Unity-based tool to integrate, benchmark, and evaluate multiple Large Language Models (LLMs)—including ChatGPT, Gemini, Claude, Mistral, and DeepSeek—for dynamic NPC dialogue

generation.Developed automated multi-agent testing scenes, real-time performance analytics dashboards (evaluating coherence, relevance, latency, and context accuracy), and dynamic voice-to-text input pipeline for interactive player-NPC conversations.Conducted prompt engineering experiments and packaged the architecture into a reusable, extensible Unity package for developers.

Demo Video – YouTube Github Repo

3. **Critter Catcher — Gameplay & AI Systems (Unity, C#):** Served as Lead Programmer for a multidisciplinary game development team, directing core gameplay mechanics and agent AI behavior systems.Architected modular, reusable C# systems to facilitate rapid iterative gameplay testing and feature prototyping. Collaborated with designers and technical artists to refine mechanics, system architecture, and overall player experience**.**
   Project Video Recording - Source Code

4. **Procedural Generation & Graphics Engine (C++, DirectX 11):** Engineered a 2.5D procedurally generated maze game using C++ and DirectX 11, focusing on high-performance graphics rendering and algorithmic content generation. Implemented custom procedural generation algorithms, collision detection engines, and real-time GPU post-processing shaders/effects.
   Demo Video and Source Code

SKILLS

Visual Studio, GitHub, Perforce, Jira, Microsoft Suite, SQL, C#, C++, Python, AWS, Azure, Google Cloud, Agile Development, Design Patterns, Data Structures, OOP, communication, collaboration, problem solving, adaptability, Time Management, Analytic, solution architect, REST, .NET, Solution Architecture Design, Requirements Analysis, Solution Presentation, Problem Solving & Decision Making, Agile / Scrum Methodologies, WebGL, SCORM.